American Computer Science League

2020-2021 ● Contest 3: Shorts ● Classroom Division

1. Boolean Algebra

Simplify the following boolean expression:

$A(\overline{B + C}) + A\overline{B}C$

A. $\overline{A}B$
B. $A\overline{B}$
C. $A + \overline{B}$
D. $\overline{AB}$
E. $\overline{A + B}$

2. Boolean Algebra

How many ordered pairs make the following expression TRUE?

$A\ (\overline{\overline{A\ +\ B}})(\overline{AB})$

A. 0
B. 1
C. 2
D. 3
E. 4

3. **Boolean Algebra**

Identify all of the ordered triples that make the following expression TRUE:

$A(\overline{B}\ + \overline{C}) + ABC$

A. (1,0,*)
B. (1,*,0)
C. (1,*,*)
D. (*,1,*)
E. (1,*,*), (0,1,1)

4. **Boolean Algebra**

How many ordered triples make the following expression TRUE?

$\overline{B}\ (\overline{A\ +\ C}) + (\overline{B\ C})$

A. 8
B. 6
C. 4
D. 2
E. 0

5. Data Structures Find the number of nodes that have only one child in the binary search tree for: SPACESHUTTLE	A. 2 B. 3 C. 4 D. 5 E. 6
6. Data Structures Given an initially empty stack and the following commands on the stack, what will the next popped item be? PUSH(H), PUSH(A), POP(X), PUSH(N), PUSH(S), POP(X), PUSH(S), POP(X), PUSH(O), PUSH(L), POP(X), PUSH(O), POP(X), POP(X)	A. L B. S C. A D. O E. N
7. **Data Structures** What is the depth of the binary search tree for: LUNARMODULE	A. 7 B. 6 C. 5 D. 4 E. 3
8. **Data Structures** Given an initially empty queue and the following commands on the queue, what will the next popped item be? PUSH(D), PUSH(A), PUSH(R), POP(X), PUSH(T), PUSH(H), POP(X), PUSH(V), PUSH(A), PUSH(D), POP(X), POP(X), PUSH(E), PUSH(R), POP(X), POP(X), POP(X)	A. A B. D C. E D. R E. nil

9. **FSAs & Regular Expressions**

Given the regular expression:

```
0(0*1U1*0)1*0*1
```

Identify all of the following strings that are **not** accepted.

A. 000010 B. 0110001 C. 0000001 D. 01110110001

E. 11111111 F. 0001111 G. 0101 H. 00001110001

A. A, B, E
B. A, B, D, H
C. A, B, E, G
D. A, E, G, H
E. A, B, E, F

10. **FSAs & Regular Expressions**

Given the regular expression:

```
[^s][aeiou][p-t]*(s|er)
```

Identify all of the following strings that are **not** accepted.

A. tater B. sorts C. faster D. plaques

E. deer F. rooster G. hits H. bursts

A. A, B, F
B. B, D, F, G
C. B, D
D. D, F, H
E. B, D, F